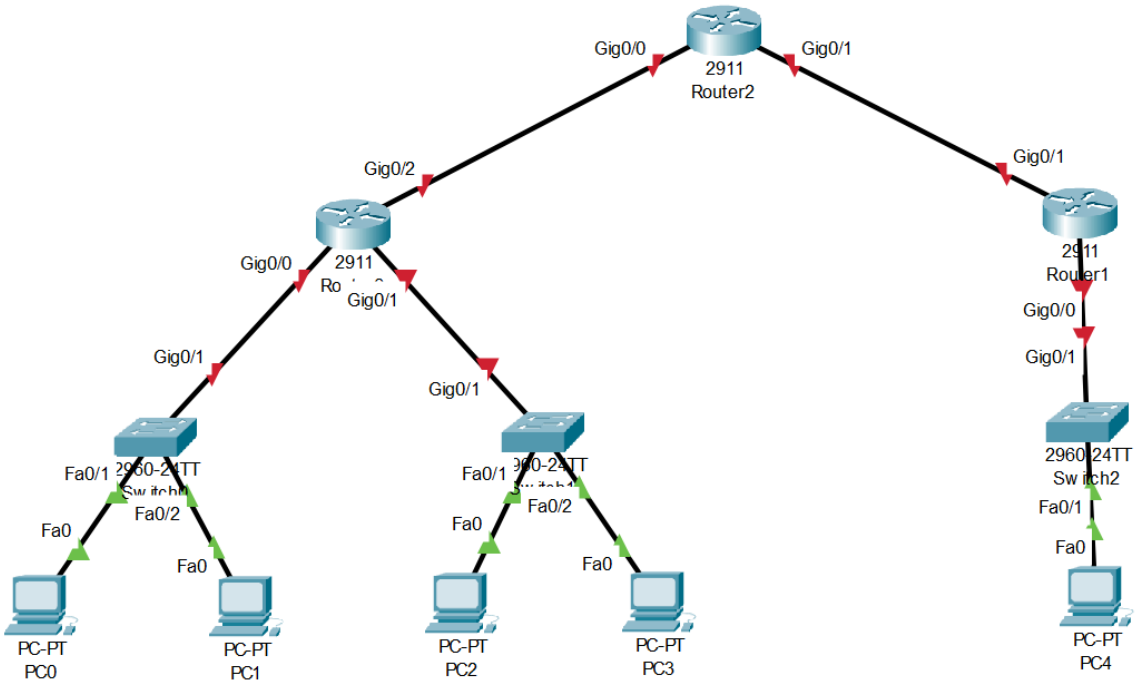
Lab 5 - Static Routing

Part 1: Network Setup

1.1 Topology and Components



In this lab, we will configure static routing using three routers where Router0 and Router1 connect to separate subnets, and Router2 serves as the backbone to facilitate communication between them. The network also includes three switches and five PCs. The goal is to establish inter-subnet communication using static routing.

1.2 IP Address Configuration

Device		IP Address	Subnet Mask	Gateway
Router 0	Gi0/0	192.168.1.1	255.255.255.0	N/A
	Gi0/1	192.168.2.1	255.255.255.0	N/A
	Gi0/2	10.0.0.1	255.255.255.252	N/A
Router 1	Gi0/0	192.168.3.1	255.255.255.0	N/A
	Gi0/1	10.0.1.1	255.255.255.252	N/A
Router 2	Gi0/0	10.0.0.2	255.255.255.252	N/A
	Gi0/1	10.0.1.2	255.255.255.252	N/A
PC0		192.168.1.2	255.255.255.0	192.168.1.1
PC1		192.168.1.3	255.255.255.0	192.168.1.1
PC2		192.168.2.2	255.255.255.0	192.168.2.1
PC3		192.168.2.3	255.255.255.0	192.168.2.1
PC4		192.168.3.2	255.255.255.0	192.168.3.1

1.2.1 Assign IP Addresses to Router Interfaces

Step 1: Go to Router0 and assign the IP address of the interface leading to the first subnet, which includes PC0 and PC1 connected through Switch0:

```
Router0>enable
Router0#configure terminal
Router0(config)#interface GigabitEthernet 0/0
Router0(config-if)#ip address 192.168.1.1 255.255.255.0
Router0(config-if)#no shutdown
Router0(config-if)#exit
```

Step 2: Now assign the IP address to the interface connected to the second subnet:

```
Router0(config)#interface GigabitEthernet 0/1
Router0(config-if)#ip address 192.168.2.1 255.255.255.0
Router0(config-if)#no shutdown
Router0(config-if)#exit
```

Step 3: Assign the IP address to the interface connected to the backbone router:

```
Router0(config)#interface GigabitEthernet 0/2
Router0(config-if)#ip address 10.0.0.1 255.255.255.252
Router0(config-if)#no shutdown
Router0(config-if)#exit
```

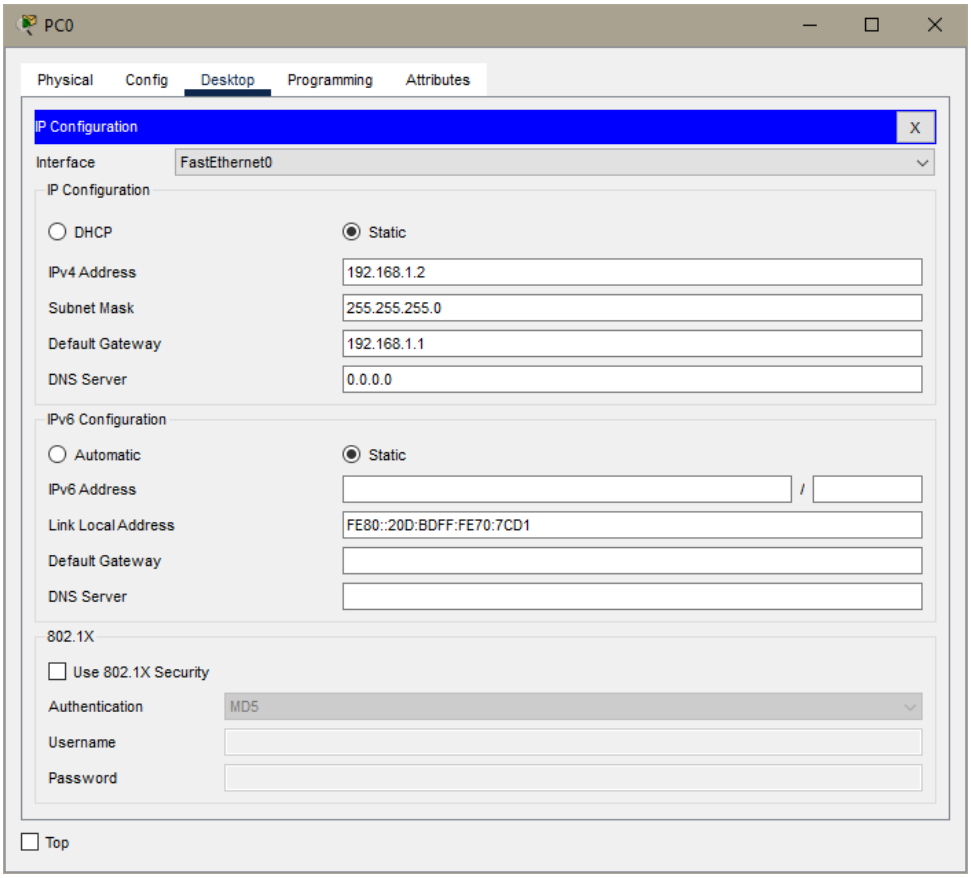
Step 4: Test the connectivity from PC0 to PC1, Router0 and PC2 using `ping` command. Was it possible to ping PC2? Why?

Step 5: Do the same for Router1 and Router2 using the table provided above

Note: Don't forget to use `no shutdown` after assigning the ip address, otherwise the interface won't communicate.

1.2.2 Assign IP Addresses to PCs

Assign IP address, subnet mask and default gateway for each PC using the table of IP addresses provided above



Part 2: Static Routing Configuration

2.1 Configure Static Routes

Step 1: On Router0, define a route to the 192.168.3.0 network via Router2:

```
Router0#configure terminal
Router0(config)#ip route 192.168.3.0 255.255.255.0 10.0.0.2
Router0(config)#exit
```

Step 2: On Router1, define routes to 192.168.1.0 and 192.168.2.0 networks via Router0:

```
Router1(config)#ip route 192.168.1.0 255.255.255.0 10.0.1.2
Router1(config)#ip route 192.168.2.0 255.255.255.0 10.0.1.2
Router1(config)#exit
```

Step 3: On Router2, define routes to networks via Router0 and Router1:

```
Router2(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router2(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.1
```

```
Router2(config)#ip route 192.168.3.0 255.255.255.0 10.0.1.1
Router2(config)#exit
```

2.2 Configure Default Routes

A default static route is used when no explicit route is found in the routing table. It ensures that packets with unknown destinations are forwarded to a next-hop router.

Step 1: Configure default static route on Router0. The only router connected to Router0 is Router2, so we use it as the next hop in the static route.

```
Router0(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.2
```

Step 2: Configure default static route on Router1 as well.

```
Router2(config)#ip route 0.0.0.0 0.0.0.0 10.0.1.2
```

Part 3: Testing and Verification

3.1 Verify Routing Tables

Use the following command on both routers to confirm the static routes:

```
Router0#show ip route
```

The result is the table of ip routes:

Router0

```
Gateway of last resort is 10.0.0.2 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/2
L       10.0.0.1/32 is directly connected, GigabitEthernet0/2
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/1
L       192.168.2.1/32 is directly connected, GigabitEthernet0/1
S    192.168.3.0/24 [1/0] via 10.0.0.2
S*   0.0.0.0/0 [1/0] via 10.0.0.2
```

Router1

```
Gateway of last resort is 10.0.1.2 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.1.0/30 is directly connected, GigabitEthernet0/1
L       10.0.1.1/32 is directly connected, GigabitEthernet0/1
S    192.168.1.0/24 [1/0] via 10.0.1.2
S    192.168.2.0/24 [1/0] via 10.0.1.2
     192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, GigabitEthernet0/0
L       192.168.3.1/32 is directly connected, GigabitEthernet0/0
S*   0.0.0.0/0 [1/0] via 10.0.1.2
```

Router2

```
Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0
C       10.0.1.0/30 is directly connected, GigabitEthernet0/1
L       10.0.1.2/32 is directly connected, GigabitEthernet0/1
S    192.168.1.0/24 [1/0] via 10.0.0.1
S    192.168.2.0/24 [1/0] via 10.0.0.1
S    192.168.3.0/24 [1/0] via 10.0.1.1
```

Ensure static routes (denoted with 'S') are listed and correctly set up.

Note the difference between the routing tables of Router2 and Router0/Router1. What does “Gateway of last resort is not set” indicate? How can you configure it?

3.2 Test Connectivity with Ping

Step 1: Check connectivity from PC0 (192.168.1.2) to PC4 (192.168.3.2) using ping command

If successful, static routing is configured correctly. If unsuccessful, check IP addresses, gateways, interface states and routing tables.

Step 2: Check connectivity from PC3 (192.168.2.3) to PC4 (192.168.3.2) using ping command

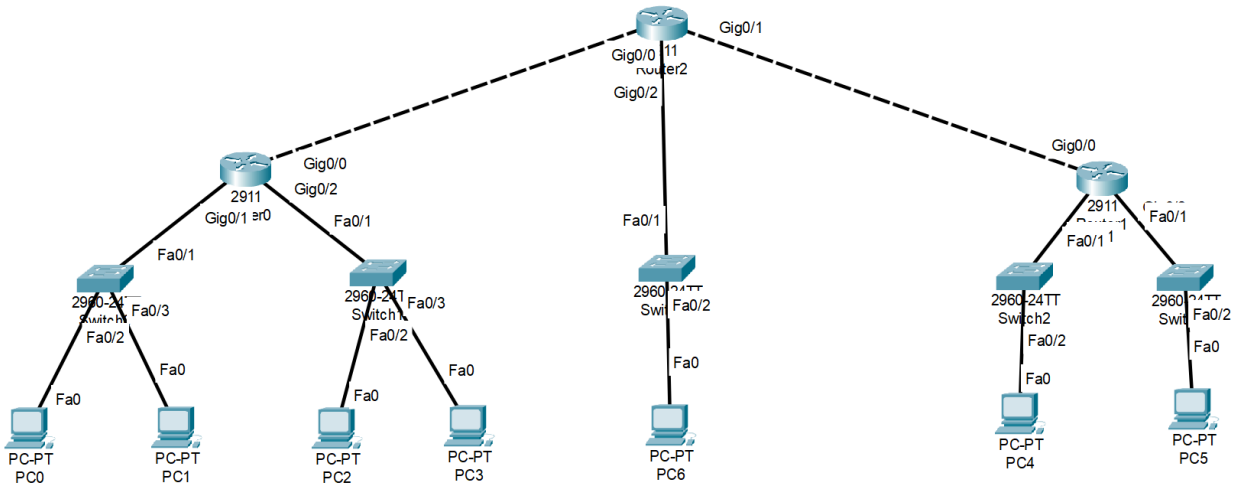
Part 4: Additional Task - Adding a New Subnet

4.1 Task Description

In this task, you will extend the network by adding new subnets connected to Router1 and Router2. Your goal is to configure the new subnets, assign appropriate IP addresses, and update the static routing configurations to ensure full network connectivity.

Steps to Complete the Task:

1. Add a new PC5 and connect it to Router1 via a new switch and PC6 to Router2 via a new switch too.



2. Choose an appropriate IP address range for these new subnets (e.g., 192.168.4.0/24).
3. Assign an IP address to the new router interface and configure the new subnet on Router1.
4. Repeat step 3 for Router2.
5. Assign an IP address and default gateway to PC5.
6. Repeat step 5 for PC6.
7. Configure static routes on Router0, Router1 and Router2 to allow full communication with the new subnet.
8. Verify connectivity by testing the routing tables and using the ping command.

Part 5: Dynamic Routing with OSPF

5.1 Task Overview

In this part, you will replace static routing with OSPF (Open Shortest Path First) dynamic routing protocol. The network topology and IP addressing remain the same. Instead of manually configuring routes, routers will automatically exchange routing information.

5.2 Remove Static Routes

Before enabling OSPF, remove all static routes from each router.

On Router0:
```
Router0(config)#no ip route 192.168.3.0 255.255.255.0 10.0.0.2
Router0(config)#no ip route 0.0.0.0 0.0.0.0 10.0.0.2
```

On Router1:
```
Router1(config)#no ip route 192.168.1.0 255.255.255.0 10.0.1.2
Router1(config)#no ip route 192.168.2.0 255.255.255.0 10.0.1.2
Router1(config)#no ip route 0.0.0.0 0.0.0.0 10.0.1.2
```

On Router2:
```
Router2(config)#no ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router2(config)#no ip route 192.168.2.0 255.255.255.0 10.0.0.1
Router2(config)#no ip route 192.168.3.0 255.255.255.0 10.0.1.1
```

5.3 Configure OSPF

Create OSPF process with process ID = 1 on each Router, assign router ID inside of OSPF process. For each Router add all directly connected networks into Area 0.
Note: Don't forget to use wildcard mask!

Router0 Configuration:
```
Router0(config)#router ospf 1
Router0(config-router)#router-id 1.1.1.1
Router0(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router0(config-router)#network 192.168.2.0 0.0.0.255 area 0
Router0(config-router)#network 10.0.0.0 0.0.0.3 area 0
Router0(config-router)#exit
```

Router1 Configuration

```
Router1(config)#router ospf 1
Router1(config-router)#router-id 2.2.2.2
Router1(config-router)#network 192.168.3.0 0.0.0.255 area 0
Router1(config-router)#network 10.0.1.0 0.0.0.3 area 0
Router1(config-router)#exit
```

Router2 Configuration

```
Router2(config)#router ospf 1
Router2(config-router)#router-id 3.3.3.3
Router2(config-router)#network 10.0.0.0 0.0.0.3 area 0
Router2(config-router)#network 10.0.1.0 0.0.0.3 area 0
Router2(config-router)#exit
```

5.4 Verification

Check OSPF neighbors:

```
show ip ospf neighbor
```

Check routing table:

```
show ip route
```

Now routes should be marked with O (instead of S for static)

5.5 Testing Connectivity

Repeat previous tests:

PC0 → PC4
PC3 → PC4

If everything is configured correctly, connectivity should work without static routes.

Compare the effort required to configure static routing vs OSPF in this topology.

Questions

1. How does a default static route differ from a manually configured static route to a specific network?
2. How does a router decide which route to use if multiple static routes exist?

3. What command would you use to remove a static route from a router?
4. How does the output of `show ip route` help in verifying static route configurations?
5. What is the main difference between static routing and OSPF?
6. What does the “O” symbol in the routing table represent?
7. Why do we use wildcard masks in OSPF instead of subnet masks?
8. What is the purpose of Area 0 in OSPF?
9. What happens if one link between routers goes down in OSPF?
10. In which cases would you prefer to use static routing, and when would you choose dynamic routing?